



# CS & IT ENGINEERING



## Algorithms

### Analysis of Algorithms

Lecture No. - 9



By- Aditya sir

# Recap of Previous Lecture



Topic

Topic

Asymptotic Notations

PYQs

TC analysis



# Topics to be Covered



Topic

Topic

Topic

TC analysis of Recursive Algo.





## About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.





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## TC analysis

1) Back-substitution

2) Master's mtd

3) Renner's Tree.

## Back-substitution

- 1) TC Recurrence
- 2) Solve to get value of Recurrence
- 3) Apply notation.

Algo  $f(n)$   $\rightarrow T(n)$

```
{  
  if (n==1) } b  
    return 1  
  
  return n * f(n-1)  
}
```

$\rightarrow T(n-1)$

$n * T(n-1)$  X  
Const



$$1) \quad T(n) = T(n-1) + a, \quad n > 1$$

$$T(n) = b, \quad n = \underline{\underline{1}}$$

2)  $T(n) = \underline{\underline{T(n-1)}} + a \text{ --- (1)}$

$T(n-1) = \underline{T(n-2) + a}$

$T(n) = T(n-2) + a + a$

$T(n) = T(n-2) + 2a \text{ --- (2)}$

$T(n-2) = T(n-3) + a$

$T(n) = T(n-3) + 3a \text{ --- (3)}$



In general:

$$T(n) = T(\underline{n-k}) + k \times a \quad \text{--- (4)}$$

Base Condition  $T(1)$

$$\hookrightarrow n-k=1$$

$$\boxed{k=n-1}$$

$$T(n) = T(1) + (n-1) \times a$$

$$\boxed{T(n) = b + (n-1) \times a} \rightarrow \text{value of Recurrence.}$$

3)

$$\boxed{T(n) = O(n)}$$



## Topic : Analysis of Algorithms

Eg.1.

Algo AJ1(n)

{

→

if (n==1)

return |

AJ2(n)

AJ1(n-1)

}

$$C1: AJ2(n) \rightarrow O(1)$$

$$C2: AJ2(n) \rightarrow O(n)$$

$$C3: AJ2(n) \rightarrow O(1/n)$$



Ques 1:  $AJ2(n) = O(1)$

$$\begin{aligned} T(n) &= T(n-1) + a, n > 1 \\ T(n) &= b, n = 1 \end{aligned}$$

done already:





$$T(n) = T(n-1) + \underline{n} \text{ --- (1)}$$

$$T(n-1) = T(n-2) + (n-1)$$

$$T(n) = \underline{T(n-2)} + \underline{n + (n-1)} \text{ --- (2)}$$

$$T(n-2) = T(n-3) + (n-2)$$

$$T(n) = T(n-3) + \underline{n + (n-1) + (n-2)} \text{ --- (3)}$$

→ generalised

→

In general,

$$T(n) = T(n-k) + \underbrace{(n + (n-1) + \dots + (n-(k-1)))}_{\text{Sum of } k \text{ terms}}$$

For Base Condition.

$$n-k=1$$

$$\boxed{k=n-1}$$



$$T(n) = T(1) + (n + (n-1) + (n-2) + \dots + \underline{\underline{(n - (n-1-1))}})$$

$$= b + (n + (n-1) + \dots + 2)$$

$$\sum_{i=2}^n i = \left[ \frac{n(n+1)}{2} - 1 \right] = \boxed{\frac{n(n-1)}{2}}$$

$$\boxed{T(n) = b + \frac{n(n-1)}{2}} \rightarrow \underline{\underline{\text{value}}}$$

$$3) \boxed{T(n) = O(n^2)} \checkmark$$

$$\begin{array}{c} 1 + 2 + \dots + n \\ 2 + 3 + \dots + n \end{array}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i = \left( \sum_{i=1}^n i \right) - 1 = \frac{n(n+1)}{2} - 1$$



Case 3 :-  $AT2(n) = O(1/n)$

$O(1/n) \rightarrow \underline{\text{decr}}$

$\text{const} > \text{decr}$

$$T(n) = T(n-1) + \underbrace{O(1/n)}_{\downarrow} + a, n > 1$$

$$T(n) = b, n = 1$$

$\rightarrow T(n) = T(n-1) + a, n > 1$

Concept:-  $T(n) = T(n-1) + \frac{1}{n} + a, n > 1$  — (1)

$$T(n-1) = T(n-2) + \frac{1}{(n-1)} + a$$

$$T(n) = T(n-2) + \frac{1}{n} + \frac{1}{n-1} + a + a$$

$$T(n) = T(n-2) + \left[ \frac{1}{n} + \frac{1}{(n-1)} \right] + 2a$$
 — (2)

$$T(n-2) = T(n-3) + \frac{1}{(n-2)} + a$$

$$T(n) = T(n-3) + \left[ \frac{1}{n} + \frac{1}{(n-1)} + \frac{1}{(n-2)} \right] + 3a \text{ --- (3)}$$

$$T(n) = T(n-k) + \left[ \frac{1}{n} + \frac{1}{(n-1)} + \dots + \frac{1}{(n-(k-1))} \right] + k \times a \text{ --- (4)}$$

for B.C,  $n-k=1$   
 $k=(n-1)$

$$T(n) = T(1) + \left[ \frac{1}{n} + \frac{1}{(n-1)} + \dots + \frac{1}{n-(n-1-1)} \right] + (n-1) \times a$$



$$T(n) = T(1) + \left[ \frac{1}{n} + \frac{1}{(n-1)} + \dots + \frac{1}{2} \right] + (n-1) \times a$$

$\left( \sum_{i=1}^n \frac{1}{i} = \log n \right)$

$\Downarrow$   
 $\log n - 1$

$$T(n) = b + (\log n - 1) + (n-1) \times a$$

3)  $T(n) = O(n)$  ✓



## Topic : Analysis of Algorithms

1. Algo AJ(n)  $\rightarrow T(n)$

```
{  
    if (n == 1)  $\left. \begin{array}{l} \text{return 1} \end{array} \right\} b$   
  
    else  
        return (AJ(n-1) + AJ(n-1))  
}
```

$\downarrow$   $\downarrow$   
 $T(n-1)$   $T(n-1)$

$\Rightarrow T(n) = b, n = 1$

$T(n) = T(n-1) + T(n-1) + a$

$T(n) = 2T(n-1) + a, n > 1$





## Topic : Analysis of Algorithms

### Step-1 Recurrence Relation

B.C  $T(n) = b, n = 1$

$$T(n) = T(n-1) + T(n-1) + a, n \geq 1$$

$$= \underline{2T(n-1)} + a, n \geq 1$$





## Topic : Analysis of Algorithms

### Step-2 Solve Recurrence

$$T(n) = 2T(n-1) + a \rightarrow \dots\dots(1)$$

$$T(n-1) = 2T(n-2) + a$$

$$T(n) = 2[2T(n-2) + a] + a \quad (2^1 - 1)a$$

$$= 2^2T(n-2) + 2a + a$$

$$= 2^2T(n-2) + 3a$$

$$T(n-2) = 2T(n-3) + a$$

$$(2^2 - 1)a$$

$$T(n) = 2^2(2T(n-3) + a) + 3a$$

$$= 2^3T(n-3) + 2^2a + 3a$$

$$= 2^3T(n-3) + 4a + 3a$$

$$= 2^3T(n-3) + 7a$$

$$= 2^3T(n-3) + (8-1)a$$

$$(2^3 - 1)a$$



## Topic : Analysis of Algorithms

### Generalized form

$$T(n) = 2^k(n-1) + (2^k - 1)a$$

For B.C.,  $(n-k) = 1$

$$k = (n-1)$$

$$T(n) = 2^{n-1} T(1) + (2^{n-1} - 1)a$$

$$= \frac{2^n}{2} \times 1 + \left(\frac{2^n}{2} - 1\right)a \rightarrow \text{value}$$

$$\Downarrow$$
$$\underline{O(2^n)}$$

$$\boxed{2^{n-1} = \frac{2^n}{2}}$$





## Topic : Analysis of Algorithms

Step-3: Apply Asymptotic Notation

$$\underline{\underline{T(n) = O(2^n)}}$$





## Topic : Analysis of Algorithms

2. Algo AJ(n)  
     $\hookrightarrow T(n)$   
{  
    if (n == 2) }  
        return 2 }  
    else  
        return AJ( $\sqrt{n}$ )  
}

$\downarrow$   
 $T(\sqrt{n})$

Challenge

A)  $O(n)$

B)  $O(\log n)$

☒ C)  $O(\log(\log n)) \rightarrow 42\%$

D)  $O(\sqrt{n}) \rightarrow 8\%$



## Topic : Analysis of Algorithms

Step.1 Find the recurrence

$$T(n) = b, \text{ } \underline{n=2}$$

$$T(n) = T(\sqrt{n}) + a, \text{ } \underline{n > 2}$$





## Topic : Analysis of Algorithms

Step.2 Solve recurrence using back substitution

$$T(n) = T(\sqrt{n}) + a$$

$$T(n) = T(n^{1/2}) + a \dots \dots (i)$$

$$T(n^{1/2}) = T(n^{1/4}) + a$$

$$T(n) = (T(n^{1/4}) + a) + a$$

$$T(n) = T(n^{1/4}) + 2a \quad \text{--- (2)}$$
$$= \cancel{T(n^{1/2})} + 2a$$

$$T(n) = T(n^{1/2^2}) + 2a$$

$$T(n) = T(n^{1/2^2}) + 2a$$

$$T(n^{1/2^2}) = T(n^{1/2^3}) + a$$

$$T(n) = (T(n^{1/8}) + 3a$$

$$T(n) = T(n^{1/2^3}) + 3a \dots \dots (2)$$
$$= \cancel{T(n^{1/2})} + 2a$$





## Topic : Analysis of Algorithms

Generalized from

$$T(n) = T(n^{1/2}) + k * a$$

For base condition  $T(2)$

$$n^{1/2^k} = 2$$

$$\frac{1}{2^k} \log_2 n = \log_2 2$$

$$2^k = \log_2 n$$

$$k = \log(\log_2 n)$$

$$T(n) = T(2) + k * a$$

$$T(n) = b + \log(\log n) * a$$
 *value*

*dominating*

Step.3- Apply Asymptotic notation

$$T(n) = O(\log(\log n))$$



## Topic : Analysis of Algorithms

#Q. Algo  $AJ(n)$

{

if ( $n==2$ )

return 2

return ( $AJ(\sqrt{n}) + AJ(\sqrt{n})$ )

}

HW



$O(n)$



$O(\sqrt{n})$



$O(\log n)$



$O(\log(\log n))$







## Topic : Analysis of Algorithms

Case-1:

Assume  $AJ2(n)$  takes constant time  $\rightarrow O(1)$





## Topic : Analysis of Algorithms

Case-2:

Assume AJ2 (n) takes constant time  $\rightarrow O(1)$



## Topic : Analysis of Algorithms

Case-3:

Assume  $AJ2(n)$  takes constant time  $\rightarrow O(1/n)$



## Topic : Analysis of Algorithms

```
1.  Algo AJ(n)
    {
        if (n == 2)
            return 2
        else
            return AJ( $\sqrt{n}$ )
    }
```





## Topic : Analysis of Algorithms

2. Algo AJ(n)

{

    if (n == 1)

        return 1

    else

        return ((AJ(n-1) + AJ (n-1)))

}



# Summary



TC analysis of Recursive Algo

Practice Questions

**THANK - YOU**